Silent Corruption in Clinical AI

At 150 dpi, 98.3% of characters are read correctly — and 15% of laboratory values are wrong

code, data & harness

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Why this matters

- Clinical AI increasingly ingests patient records as **images** — scans, faxes, phone photographs — not structured data.
- Every stage from pixel to generated sentence is lossy: OCR, extraction, abnormal-value flagging, prompt assembly.
- RAG systems are evaluated on retrieval and answer quality — almost never on whether the patient's own numbers survived.
- A wrong but still *plausible* value crosses the same threshold as the truth, so the alarm the clinician sees is identical.
- If nothing can say “this reading is suspect”, a decimal point lost in a scan becomes a clinical recommendation.

What I did

- Built a RAG clinical decision support prototype: React, FastAPI, Qdrant, Tesseract OCR, LLM generation. Code is public.
- Injected **controlled image faults** into a synthetic discharge summary with machine-readable ground truth for 13 analytes.
- 32 conditions — resolution, blur, JPEG, rotation, noise, composites — 5 seeds where stochastic. **832 observations.**
- Added a **calibration gate**: no result unless the clean render gives 13/13. My first run failed it — my own parser's bug.
- Measured at three levels: characters, extracted values against truth, and the flag the clinician would be shown.

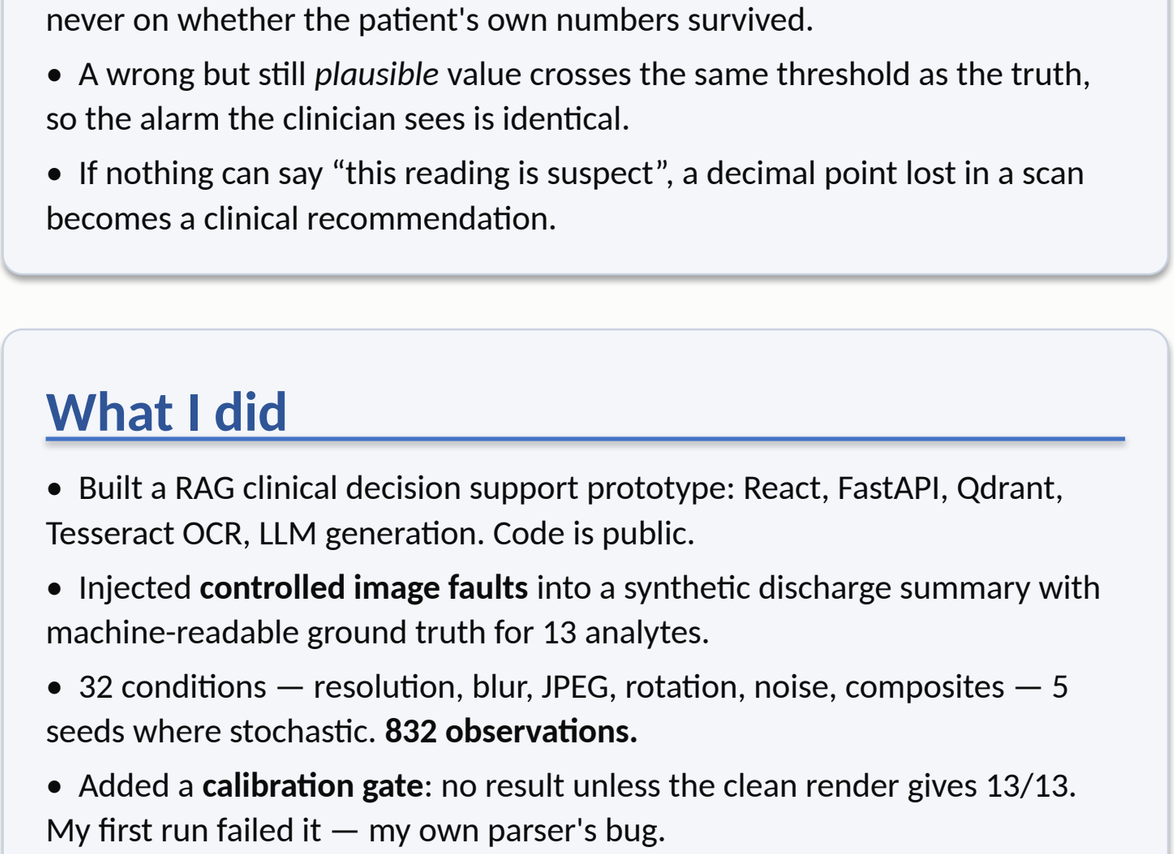


Figure 2. Error at two levels against scan resolution; both are fractions on one axis. 13 analytes per point. Blur, JPEG, rotation and noise at 300 dpi were each harmless until a cliff (see repository).

What this does not show

One synthetic document, 13 analytes, one layout, one OCR engine (Tesseract 5.5.1). Degradations are synthetic approximations, not real scans. Extraction used a deterministic row parser; the deployed LLM extractor is untested. Flags follow the system's own hardcoded ranges, which are not age- or sex-aware. Plausibility envelopes are my own construction, not a clinical standard. Generation used one 3B local model, six samples per condition. No clinician has judged whether any corruption would change management. This is a preliminary fault-injection study on a research prototype — not a validated or deployable system.

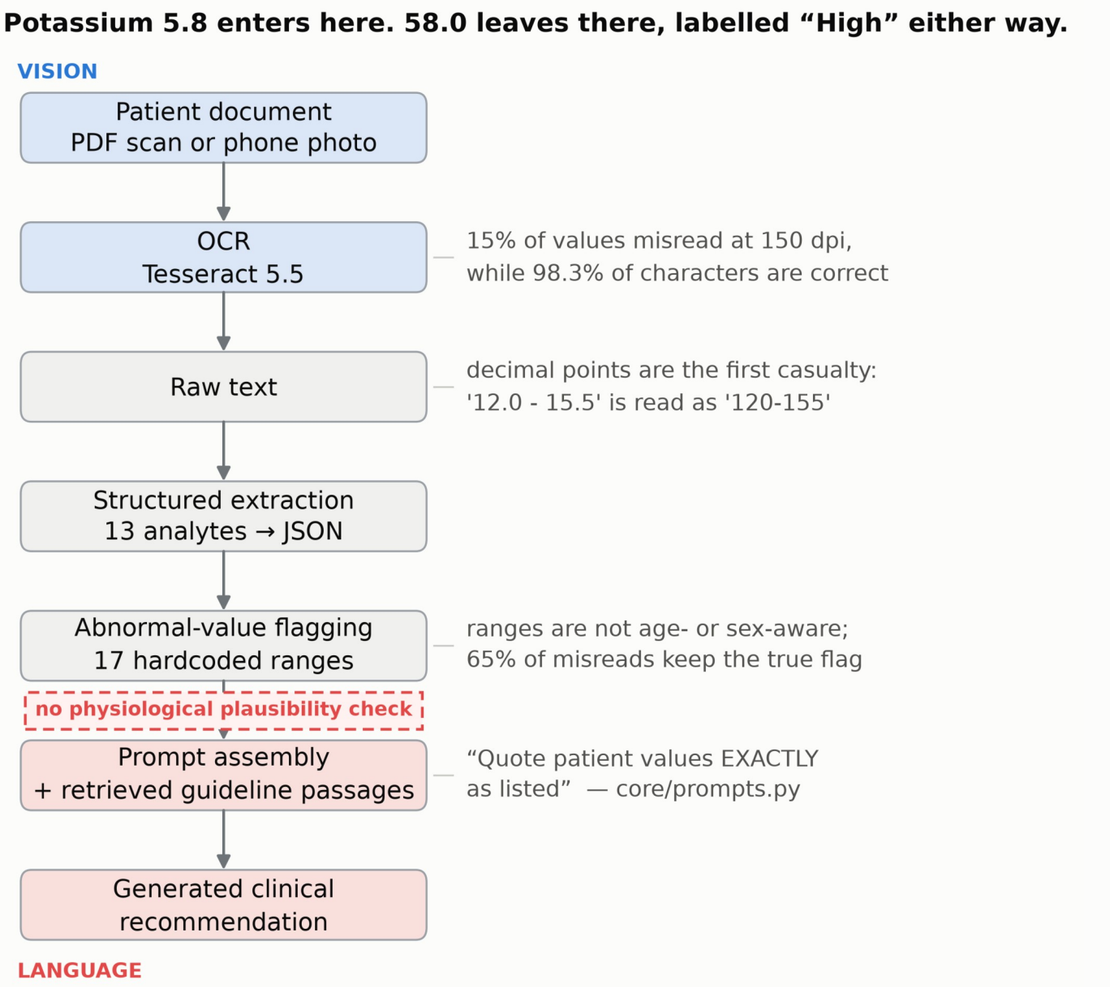


Figure 1. The measured pipeline. Vision at the top, language at the bottom, five lossy stages between — and no check that a value is physiologically possible.

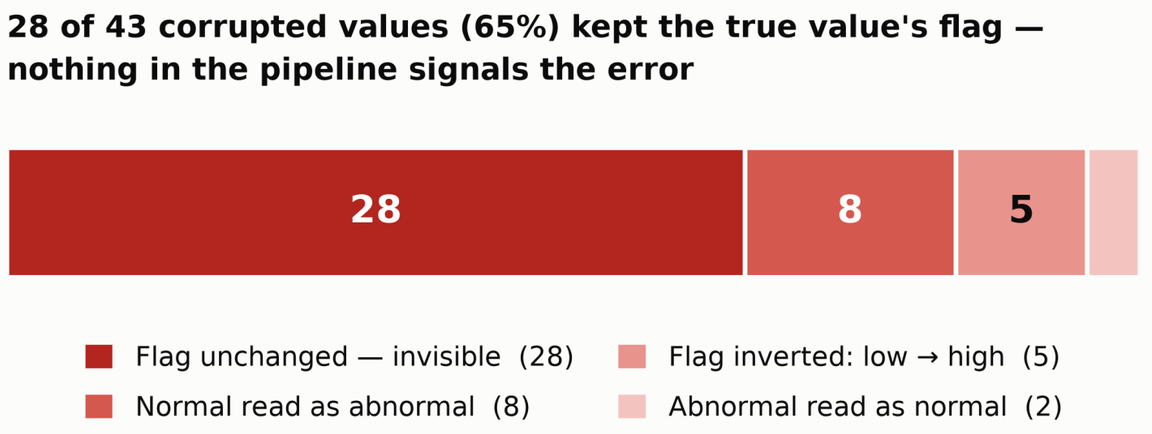


Figure 3. What the flagging layer did with each of the 43 corrupted values.

What I found

1. Text quality does not predict value safety.
At 150 dpi — a routine scanner setting — 98.3% of characters are correct and 15% of values are wrong. Resolution dominates: at 300 dpi, blur to σ 2.0, JPEG to q15, rotation to 5° and noise to σ 20 were each harmless.

2. 65% of corruptions are invisible.
Of 43 wrong values, 28 produced the **same** abnormal-value flag as the truth. Potassium 5.8 → 58.0 mEq/L is still “High”; troponin 0.07 → 7.0 is still “Critical”. 60% were pure decimal-place shifts.

3. It reaches the generated output.
The corrupted value is quoted verbatim in the generated recommendation in 5 of 6 generations at 100 dpi and 4 of 6 at 150 dpi — the prompt instructs the model to quote values exactly. Whether the *recommendation* itself changes was **not stable** across generations; that needs a larger, properly powered comparison.

Grounding tells you where a number came from. It does not tell you whether it was read correctly.

References

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